Fig 7th NRN

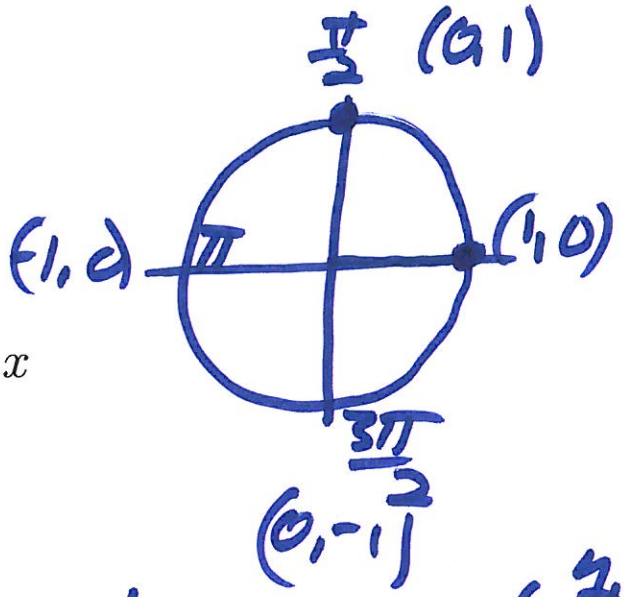
Lecture 28: Section 4.5
Graphs of Sine and Cosine Functions

Period of sine and cosine

Amplitude of sine and cosine

Horizontal translation - phase shift

Vertical translations



angle

output

The Graph of $y = \sin x$

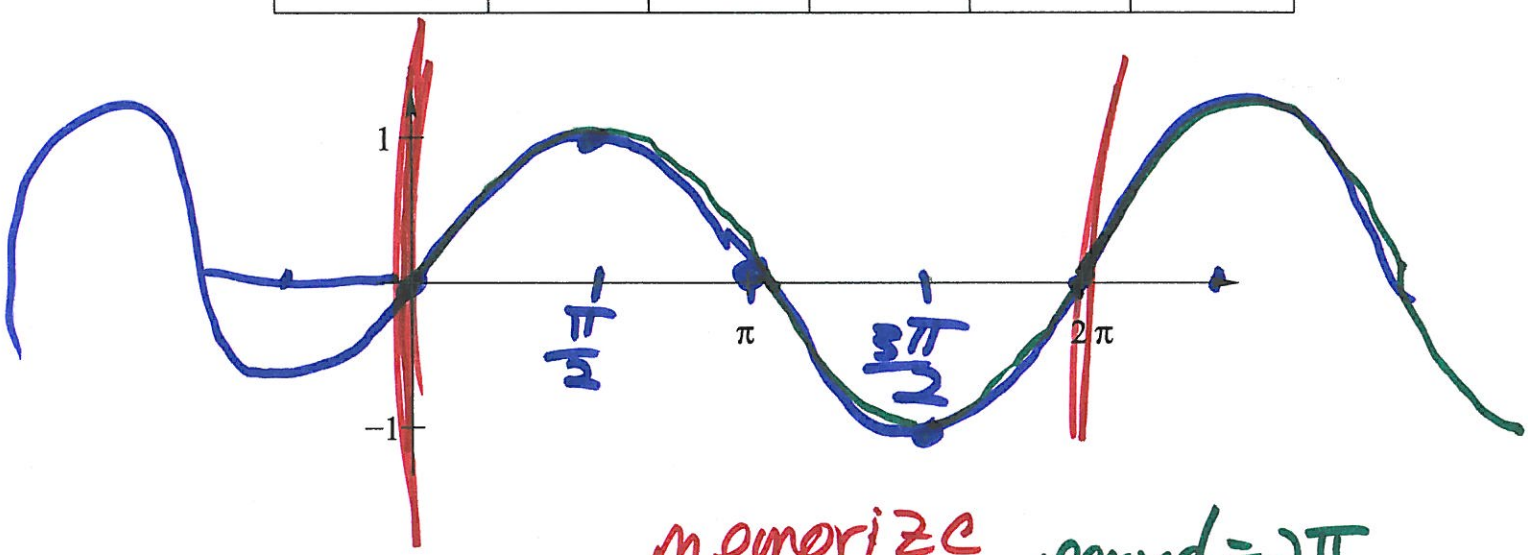
Domain: $(-\infty, \infty)$

Range: $[-1, 1]$

*Period: 2π starts repeating the cycle (values)

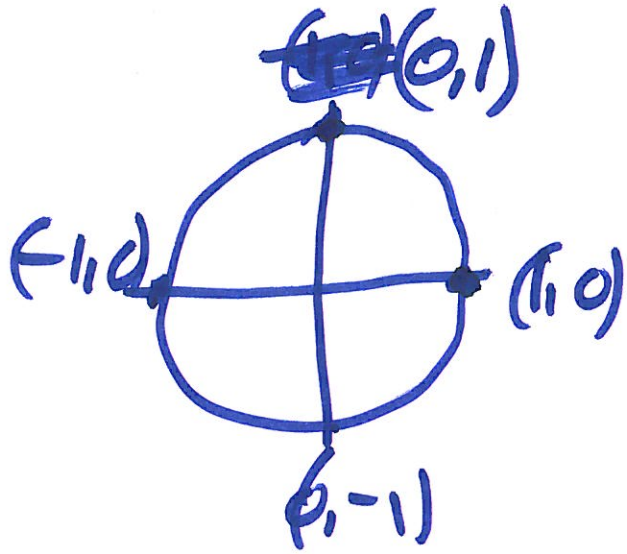
The key points in one period:

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \sin x$	0	1	0	-1	0



memorize

period $= 2\pi$



The Graph of $y = \cos x$

Domain: $(-\infty, \infty)$

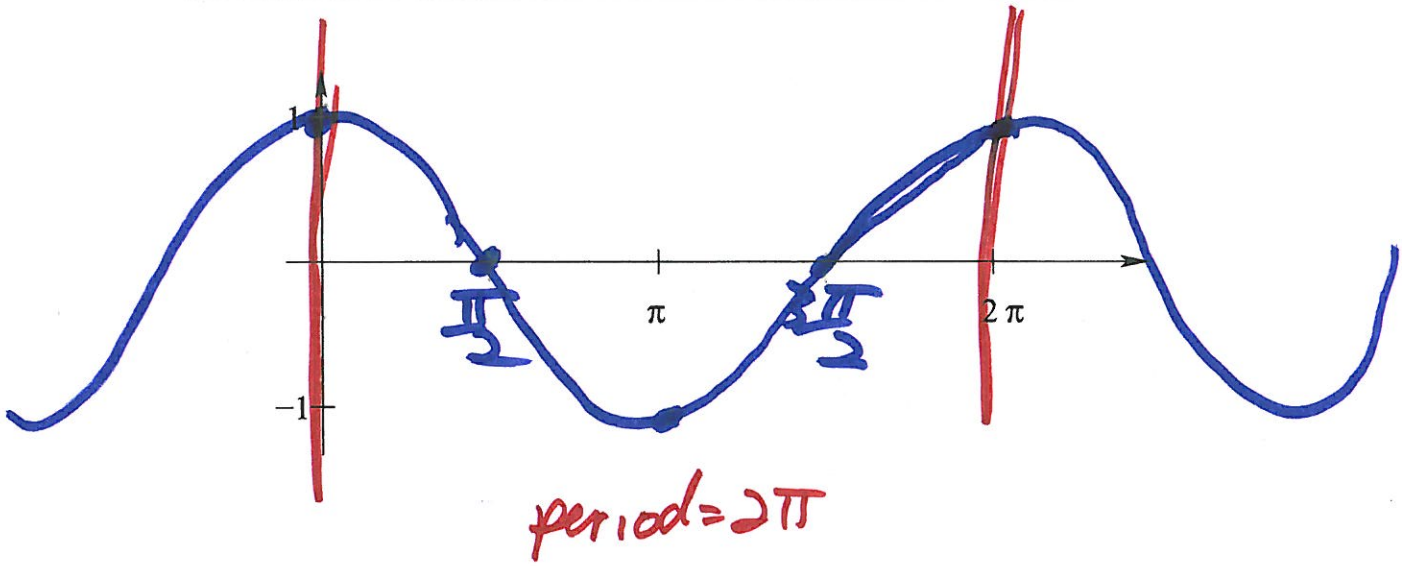
Range: $[-1, 1]$

Period: 2π

The key points in one period:

cos θ = x-value on the unit circle.

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \cos x$	1	0	-1	0	1



Amplitude for Functions $y = a\sin x$ and $y = a\cos x$

Def. The **amplitude** of $y = a\sin x$ and $y = a\cos x$ represents half the distance between the maximum and minimum values of the function and is given by

$$\text{Amplitude} = |a|$$

$$= \frac{\text{maximum value - minimum value}}{2}$$

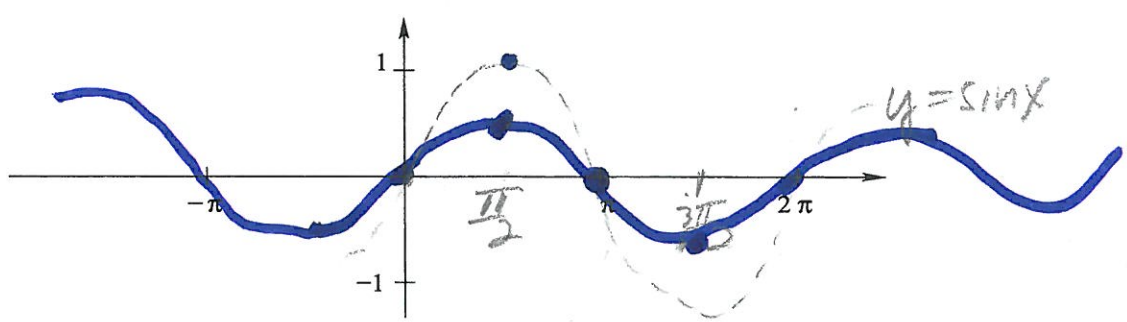
NOTE:

1. If $|a| > 1$, the curve is stretched vertically, and if $|a| < 1$, the curve is shrunk vertically.

2. The range is $[-a, a]$.

$y = 7\sin x$

ex. Graph $y = \frac{1}{2}\sin x$ $A = \left|\frac{1}{2}\right| = \frac{1}{2}$ $R: \left[-\frac{1}{2}, \frac{1}{2}\right]$



$$\frac{\frac{1}{2} - \frac{-1}{2}}{2} = \frac{1}{2}$$

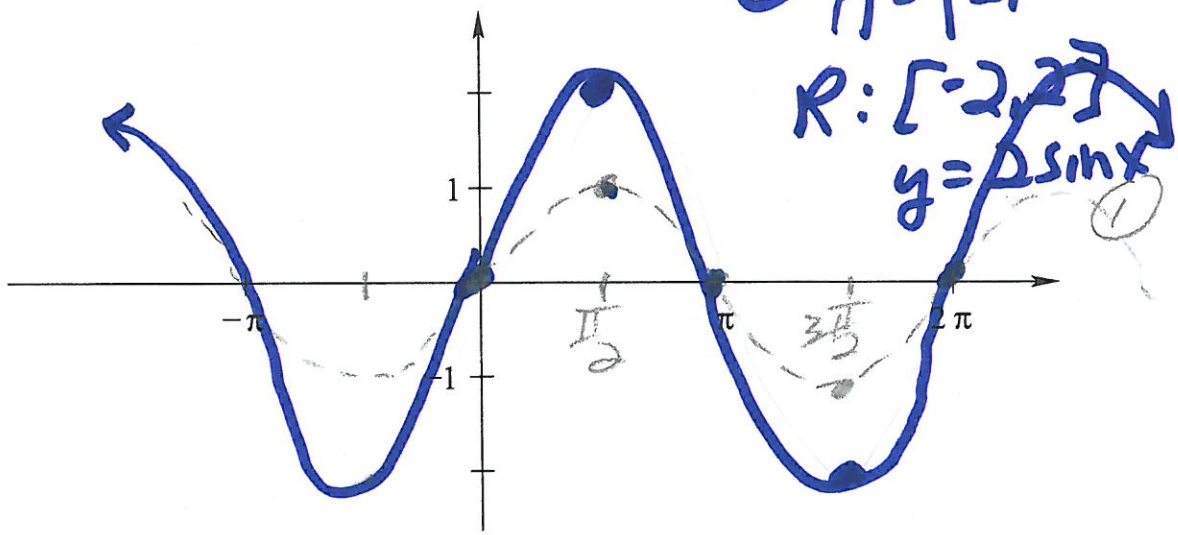
ex. Graph $y = 2\sin x$

① $y = \sin x$

② $A = |2| = 2$

$R: [-2, 2]$

$y = 2\sin x$



ex. Graph $y = -2\cos x$

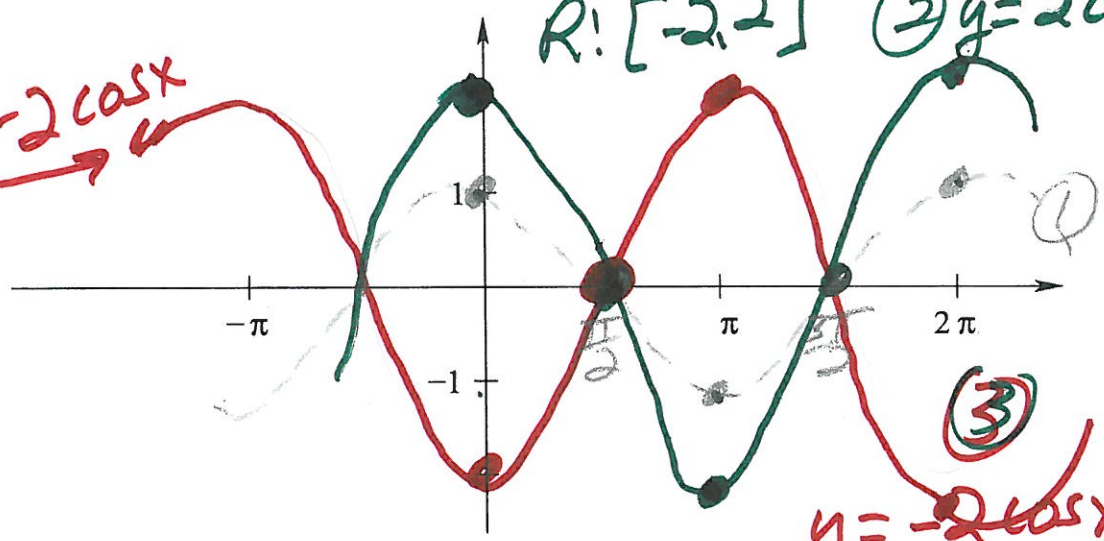
$A = |-2| = 2$

$R: [-2, 2]$

① $y = \cos x$

② $y = 2\cos x$

$y = -2\cos x$



③ $y = -2\cos x$

refl. across the x-axis

$y = \sin x$ has normal period from $0 - 2\pi$

period on $y = a\sin(bx)$ is: set $bx = 0 \rightarrow x = 0$

set $bx = 2\pi \rightarrow x = \frac{2\pi}{b}$

Period for Functions $y = a\sin(bx)$ and $y = a\cos(bx)$

how long is the cycle?

The **period** of $y = a\sin(bx)$ and $y = a\cos(bx)$ is given by

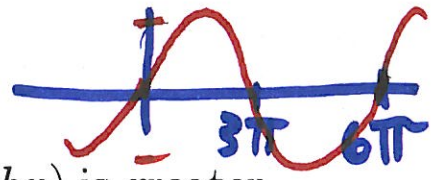
$y = \sin(\frac{1}{3}x)$

$$\text{Period} = \frac{2\pi}{b}$$

$\frac{1}{3}x = 0 \rightarrow x = 0$

where b is a positive real number.

$\frac{1}{3}x = 2\pi \rightarrow x = 6\pi$



NOTE:

1. If $0 < b < 1$, the period of $y = a\sin(bx)$ is greater than 2π and represents a horizontal stretching of the graph $y = a\sin x$.

2. If $b > 1$, the period of $y = a\sin(bx)$ is less than 2π and represents a horizontal shrinking of the graph $y = a\sin x$.

ex. Find the amplitude and period of each function, and sketch its graph.

$A = |4| = 4$

period $= \frac{2\pi}{b} \rightarrow \frac{2\pi}{3}$

horizontal shrink.

vertical stretch $\times 4$

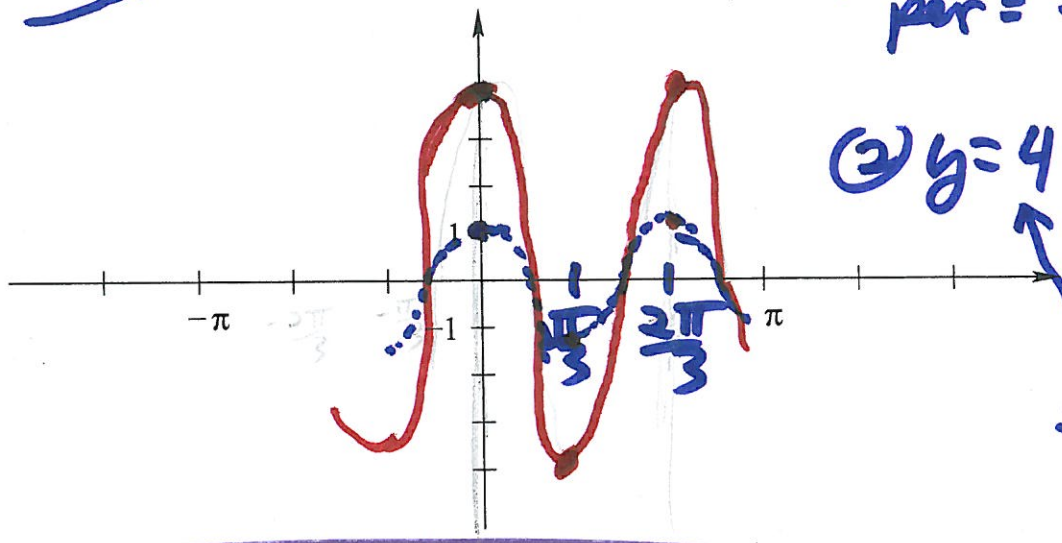
1) $y = 4\cos(3x)$

① $y = \cos 3x$

per $= \frac{2\pi}{3}$

② $y = 4\cos(3x)$

vert stretch $\times 4$



or $y = 4\cos(3x)$

$3x = 0 \rightarrow x = 0$

$3x = 2\pi \rightarrow x = \frac{2\pi}{3}$

x	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$
y	4	0	-4	0	4

1 0 −1 0 1

$\frac{1}{2}\left(0 + \frac{2\pi}{3}\right) = \frac{2\pi}{6} = \frac{\pi}{3}$

$\frac{1}{2}\left(0 + \frac{\pi}{3}\right) = \frac{\pi}{6}$

$\frac{1}{2}\left(\frac{\pi}{3} + \frac{2\pi}{3}\right) = \frac{1}{2}\left(\frac{3\pi}{3}\right) = \frac{\pi}{2}$

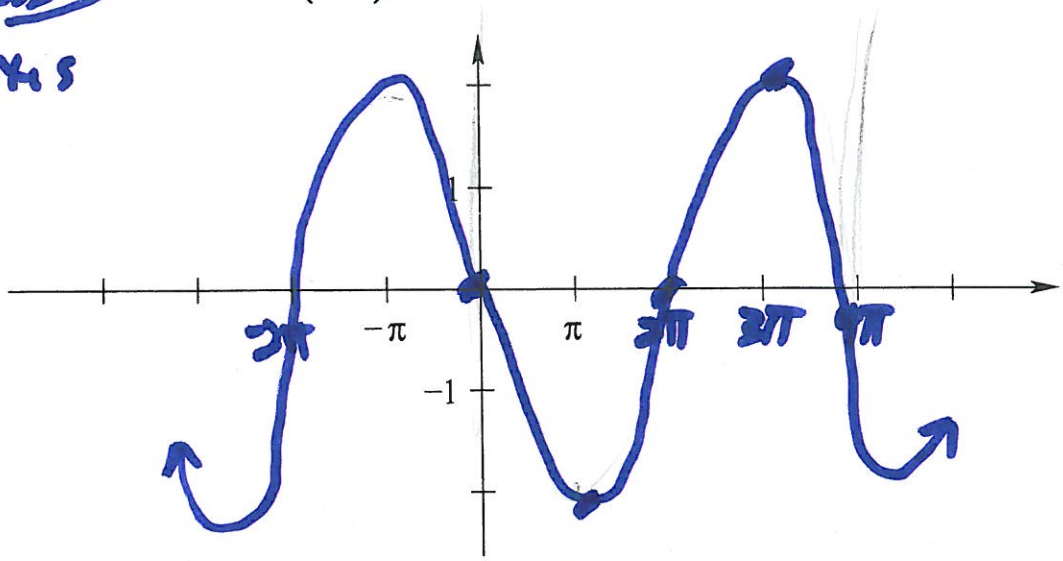
$A = |-2| = 2$

period $= \frac{2\pi}{b} \rightarrow \frac{2\pi}{\frac{1}{2}} = 4\pi = \text{per}$

horizontal stretch

2) $y = -2\sin\left(\frac{1}{2}x\right)$

refl. across the x-axis



or

$\frac{1}{2}x = 0 \rightarrow x = 0$

$\frac{1}{2}x = 2\pi \rightarrow x = 4\pi$

x	0	π	2π	3π	4π
y	0	-2	0	2	0
y = sin x	0	1	0	-1	0

$y = -2\sin\left(\frac{1}{2}x\right)$

Horizontal Translations

	$y = a\sin(bx)$	$y = a\sin(bx - c)$
Amplitude	$A = \lvert a \rvert$	$A = \lvert a \rvert$
Period	$P = \frac{2\pi}{b}$	$P = \frac{2\pi}{b}$
One cycle	$0 \le bx \le 2\pi$ $0 \le x \le \frac{2\pi}{b}$	$0 \le bx - c \le 2\pi$ $\frac{c}{b} \le x \le \frac{2\pi}{b} + \frac{c}{b}$

NOTE: The graph of $y = a\sin(bx)$ is shifted by $\frac{c}{b}$ units to get the graph of $y = a\sin(bx - c)$. The number $\frac{c}{b}$ is called the **phase shift**.

minus

ex. Find the amplitude, period and phase shift of each function, and sketch its graph.

$A = \left|\frac{3}{2}\right| = \frac{3}{2}$

$p.s. = \frac{c}{b} = \frac{\frac{\pi}{2}}{2} = \frac{\pi}{4} = p.s.$

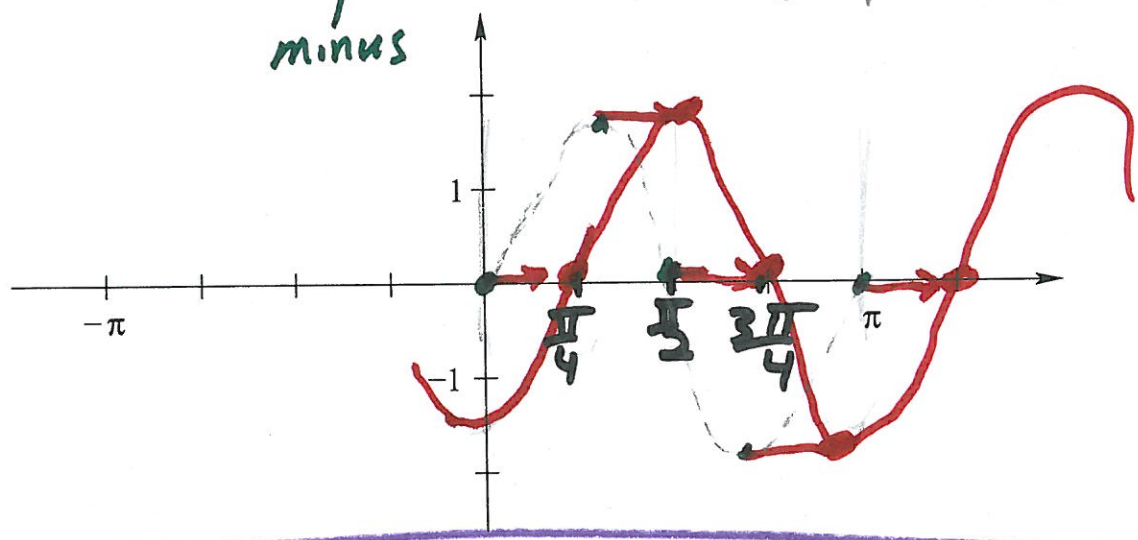
period $= \frac{2\pi}{b} \rightarrow \frac{2\pi}{2} = \pi$

① $y = \frac{3}{2}\sin(2x)$

② $p.s. = \frac{\pi}{4}$

1) $y = \frac{3}{2}\sin\left(2x - \frac{\pi}{2}\right)$

minus



or

$2x - \frac{\pi}{2} = 0 \rightarrow x = \frac{\pi}{4}$

$2x - \frac{\pi}{2} = 2\pi \rightarrow x = \frac{5\pi}{4}$

x	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$
y	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	0
	0	1	0	−1	0

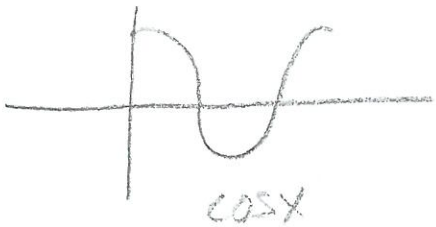
$y = \frac{3}{2}\sin\left(2x - \frac{\pi}{2}\right)$

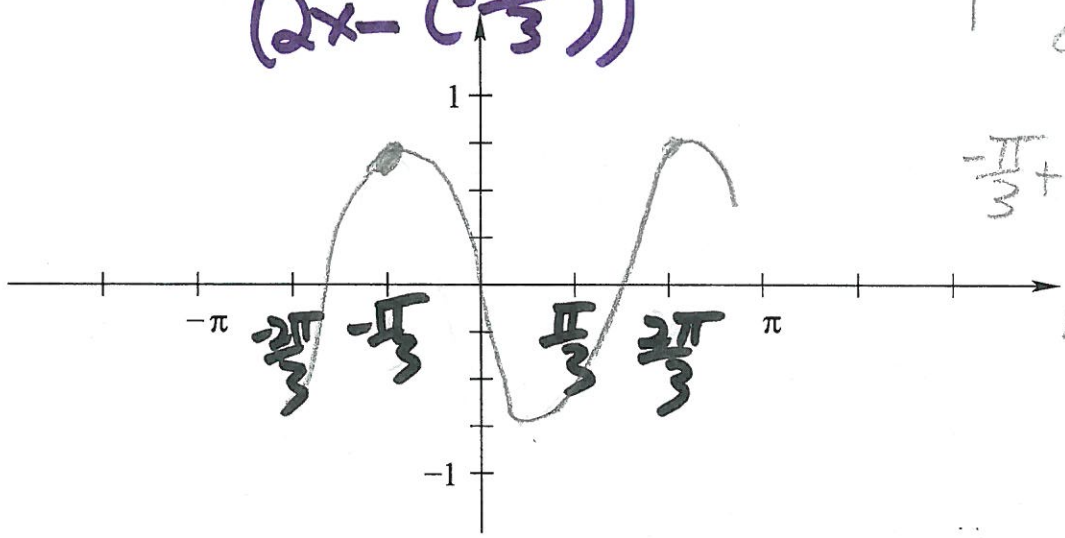
$A = \left|\frac{3}{4}\right| = \frac{3}{4}$ $\quad$ $p.s. = \frac{c}{b} \rightarrow \frac{-\frac{2\pi}{3}}{2} = -\frac{\pi}{3}$

$Per = \frac{2\pi}{2} = \pi$

2) $y = \frac{3}{4}\cos\left(2x + \frac{2\pi}{3}\right)$

$\left(2x - \left(-\frac{2\pi}{3}\right)\right)$





$-\frac{\pi}{3} + \pi = \frac{2\pi}{3}$

↑

Period

as a check ↓

x				
y				

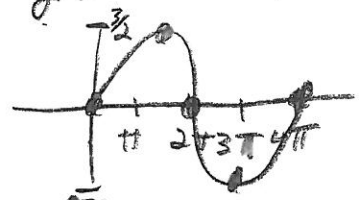
Graphing by using period and phase shift.

$A = \left|\frac{3}{2}\right| = \frac{3}{2}$

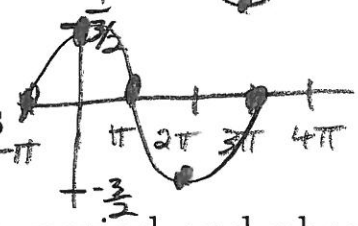
$per = \frac{2\pi}{b} = \frac{2\pi}{\frac{1}{2}} = 4\pi$

$p.s. = \frac{c}{b} = \frac{-\frac{\pi}{2}}{\frac{1}{2}} = -\pi$

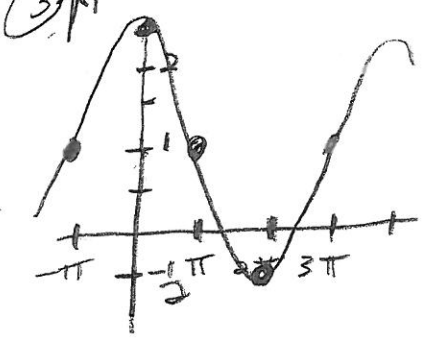
① per = 4π



② p.s. = $-\pi$



③ ↑1



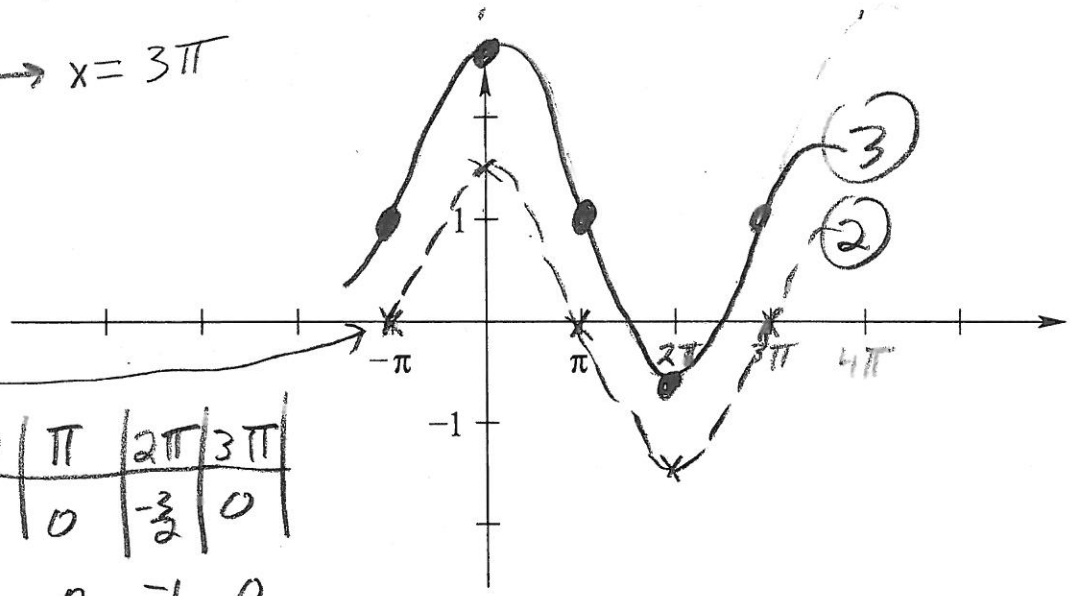
Vertical Translations

ex. Find the amplitude, period and phase shift of $y = 1 + \frac{3}{2}\sin\left(\frac{x}{2} + \frac{\pi}{2}\right)$, and sketch its graph.

The above three steps using period and phase shift translations are the quickest. Below is the other approach to graphing that I have been using.

① $\frac{x}{2} + \frac{\pi}{2} = 0 \rightarrow x = -\pi$

$\frac{x}{2} + \frac{\pi}{2} = 2\pi \rightarrow x = 3\pi$



②

x	$-\pi$	0	π	2π	3π
y	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	0

$y = \sin x$ 0 1 0 −1 0

③ ↑1

x	$-\pi$	0	π	2π	3π
y	1	$\frac{5}{2}$	1	$-\frac{1}{2}$	1

ex. As a wave passes by an offshore piling, the height of the water is modeled by the function

$$h(t) = 3\cos\left(\frac{\pi}{10}t\right)$$

where $h(t)$ is the height in feet above mean sea level at time t seconds.

1) Find the period of the wave. $\frac{2\pi}{b} \rightarrow \frac{2\pi}{\frac{\pi}{10}} = 20$ seconds

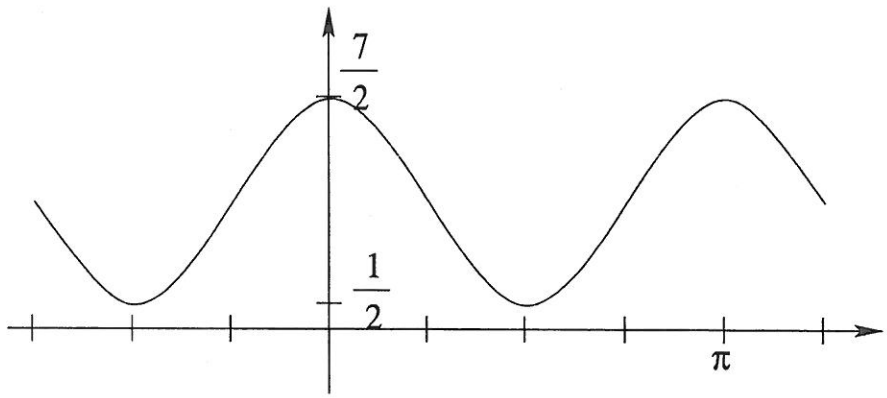
2) Find the wave height, that is, the vertical distance between the trough and the crest of the wave.

Find amplitude and double it.

$3 \times 2 = 6$ feet

Practice.

Find at least two equations for the graph.



Answer. $y = \frac{3}{2}\sin\left(2x + \frac{\pi}{2}\right) + 2,\ y = \frac{3}{2}\cos(2x) + 2,$

$y = -\frac{3}{2}\sin\left(2x - \frac{\pi}{2}\right) + 2,\ y = -\frac{3}{2}\cos(2x + \pi) + 2, \cdots$